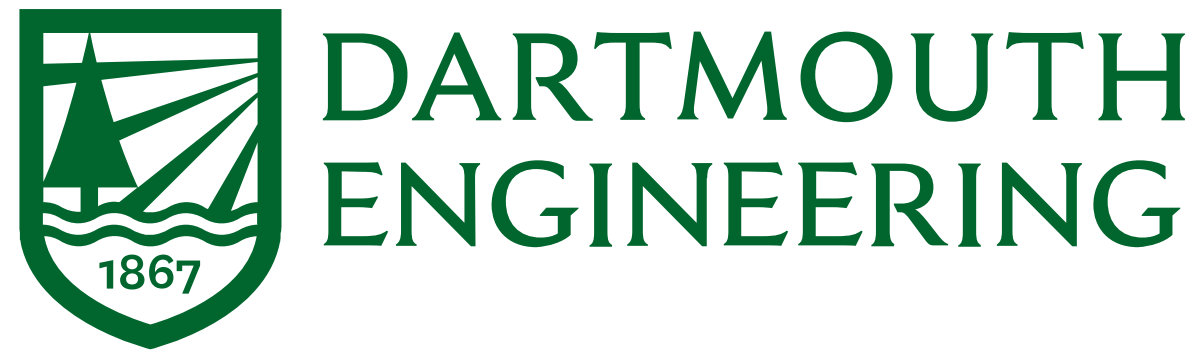


Simultaneous Illumination Normalization and Sparse Image Compression via Unfolded Recovery

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Motivation

Cameras sample then throw away. A JPEG keeps only 5–10% of DCT coefficients, and the ISP further reshapes the data through demosaicing, tone-mapping, and white-balance. Compressed sensing reverses this: $M \ll N$ random measurements suffice if the signal is sparse [? ?].

The sequential pipeline wastes its prior. RAW sensors measure $\boldsymbol{x} = \boldsymbol{s} \odot \boldsymbol{g}$, the *product* of sparse reflectance and a smooth illumination field. CS-then-correct enforces ℓ_1 on $\boldsymbol{x}$, but $\boldsymbol{x}$ is *not* DCT-sparse; multiplication by $\boldsymbol{g}$ destroys the sparsity the regularizer relies on. Post-hoc column-mean correction can rescale but cannot restore information already shrunk away.

Our Contributions

Joint recovery objective. A single ℓ_1 +smooth-prior problem in $(\boldsymbol{c}, \boldsymbol{g})$ solved by block-coordinate alternation: FISTA on $\boldsymbol{c}$, closed-form ridge on $\boldsymbol{g}$.

Deep-unfolded inner solver. A K =10-layer LISTA [?] with Adam, gradient clipping and best-checkpoint restore for stability.

Scene-dependent benchmark. Phase transition, multi-image rate-distortion, LISTA-vs-classical, and per-scene illumination ablation on 5 natural images.

Honest failure-mode analysis. Joint catastrophically fails on near-uniform scenes (moon, −8 dB); we trace it to rank-deficiency of $\boldsymbol{B}^\top \boldsymbol{B}$.

Big Takeaway

Estimating reflectance and illumination *jointly* beats the conventional CS-then-correct pipeline by up to +4.7 dB on textured scenes – *when* the scene has enough column-wise energy to identify the gain.

Textured wins of +1.1 to +4.7 dB on cameraman and astronaut; **LISTA speedup** where K =10 layers match FISTA at 22 iterations; **failure mode** of −8 dB on near-uniform moon – exposing the need for an identifiability gate.

Background: Compressed Sensing

With $\boldsymbol{x} = \boldsymbol{\Psi} \boldsymbol{c}$ and $\|\boldsymbol{c}\|_0 \leq S$, the forward model is $\boldsymbol{y} = \boldsymbol{A} \boldsymbol{x} + \boldsymbol{w}$, $\boldsymbol{A} \in \mathbb{R}^{M \times N}$. If $\boldsymbol{A}$ satisfies the RIP with $\delta_{2S} < \sqrt{2} - 1$, the Lasso relaxation

$$\boldsymbol{c}_\lambda = \arg \min_{\boldsymbol{c}} \frac{1}{2} \|\boldsymbol{A} \boldsymbol{\Psi} \boldsymbol{c} - \boldsymbol{y}\|_2^2 + \lambda \|\boldsymbol{c}\|_1$$

recovers $\boldsymbol{c}$ to within $O(\epsilon)$ [?]. We use the 2D DCT as $\boldsymbol{\Psi}$, a strong natural-image sparsifying basis.

Block-Coordinate Recipe

Initialise $\boldsymbol{g}^{(0)} = \mathbf{1}$ and $\boldsymbol{c}^{(0)} = \mathbf{0}$. The **c-update** holds $\boldsymbol{g}^{(t)}$ fixed and runs T_c FISTA (or K LISTA) steps on the effective sensing matrix $\boldsymbol{A}_{\text{eff}}(\boldsymbol{g}^{(t)})$. The **g-update** then holds $\boldsymbol{c}^{(t+1)}$ fixed and solves $(\boldsymbol{B}^\top \boldsymbol{B} + \lambda_g \boldsymbol{L}) \boldsymbol{g} = \boldsymbol{B}^\top \boldsymbol{y}$ in closed form, clamping to $[\epsilon_g, \infty)$. We repeat for $T_{\text{outer}}=8$ outer iterations and resolve the scalar gain ambiguity at evaluation.

Encoder → Decoder Pipeline

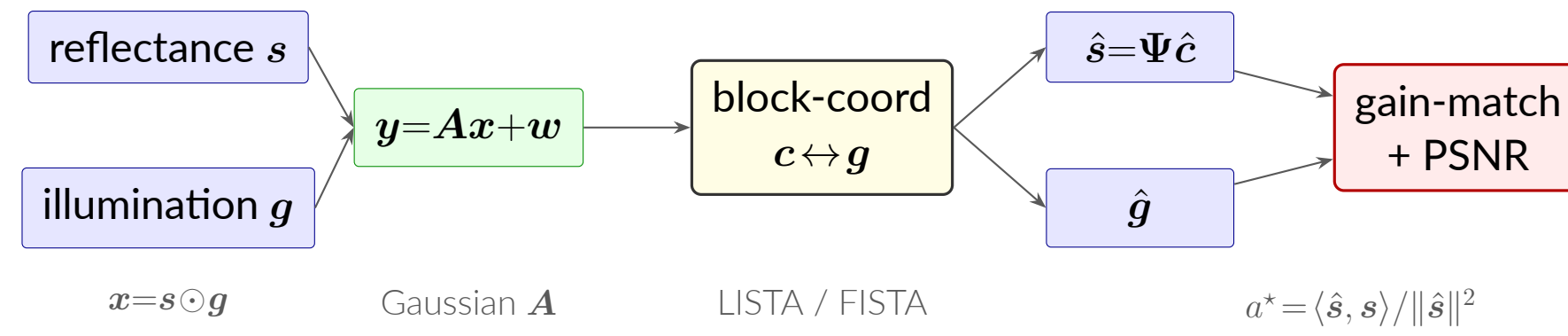


Figure 1. End-to-end joint recovery pipeline. The RAW signal $\boldsymbol{x} = \boldsymbol{s} \odot \boldsymbol{g}$ is measured via Gaussian $\boldsymbol{A}$; block-coordinate alternation recovers $(\boldsymbol{c}, \boldsymbol{g})$ simultaneously; the scalar gain ambiguity is resolved at evaluation.

Method 1: Joint Recovery Objective

Replace the two-stage pipeline with a single objective:

$$\min_{\boldsymbol{c}, \boldsymbol{g}} \frac{1}{2} \left\| \boldsymbol{A} \text{diag}(\boldsymbol{g}_{\text{pix}}) \boldsymbol{\Psi} \boldsymbol{c} - \boldsymbol{y} \right\|_2^2 + \lambda_c \|\boldsymbol{c}\|_1 + \lambda_g \|\nabla \boldsymbol{g}\|_2^2$$

Jointly non-convex but **block-convex** in each variable.

c-step. Fix $\boldsymbol{g}$; effective sensing matrix $\boldsymbol{A}_{\text{eff}} = \boldsymbol{A} \text{diag}(\boldsymbol{g}_{\text{pix}}) \boldsymbol{\Psi}$, update is a standard Lasso solved by FISTA.

g-step. Fix $\boldsymbol{c}$; with $\hat{\boldsymbol{s}} = \boldsymbol{\Psi} \boldsymbol{c}$ the model is linear in $\boldsymbol{g}$, $\boldsymbol{y} = \boldsymbol{B}(\hat{\boldsymbol{s}}) \boldsymbol{g} + \boldsymbol{w}$, and the update is closed-form ridge:

$$\boldsymbol{g}^* = (\boldsymbol{B}^\top \boldsymbol{B} + \lambda_g \boldsymbol{L})^{-1} \boldsymbol{B}^\top \boldsymbol{y},$$

with $\boldsymbol{L} = \nabla^\top \nabla$ the 1D discrete Laplacian. We clamp $\boldsymbol{g} \geq \epsilon_g$ to avoid divide-by-zero.

Gain ambiguity. The model has scalar invariance $(\boldsymbol{s}, \boldsymbol{g}) \rightarrow (a \boldsymbol{s}, \boldsymbol{g}/a)$; we resolve at evaluation by the optimal positive gain $\alpha^* = \langle \hat{\boldsymbol{s}}, \boldsymbol{s}_{\text{true}} \rangle / \|\hat{\boldsymbol{s}}\|_2$, applied identically to baselines.

Method 2: Deep-Unfolded LISTA Inner Solver

Replace the FISTA **c**-step with K learned layers [?]:

$$\boldsymbol{c}^{(k+1)} = \mathcal{S}_{\theta_k}(\boldsymbol{W}_t \boldsymbol{c}^{(k)} + \boldsymbol{W}_c \boldsymbol{y}), \quad k = 0, \dots, K-1.$$

Initialise $\boldsymbol{W}_c = \frac{1}{L}(\boldsymbol{A} \boldsymbol{\Psi})^\top$, $\boldsymbol{W}_t = \boldsymbol{I} - \frac{1}{L}(\boldsymbol{A} \boldsymbol{\Psi})^\top \boldsymbol{A} \boldsymbol{\Psi}$, $\theta_k = \lambda/L$ to reproduce ISTA at step zero, then train end-to-end on $(\boldsymbol{y}, \boldsymbol{c})$ pairs.

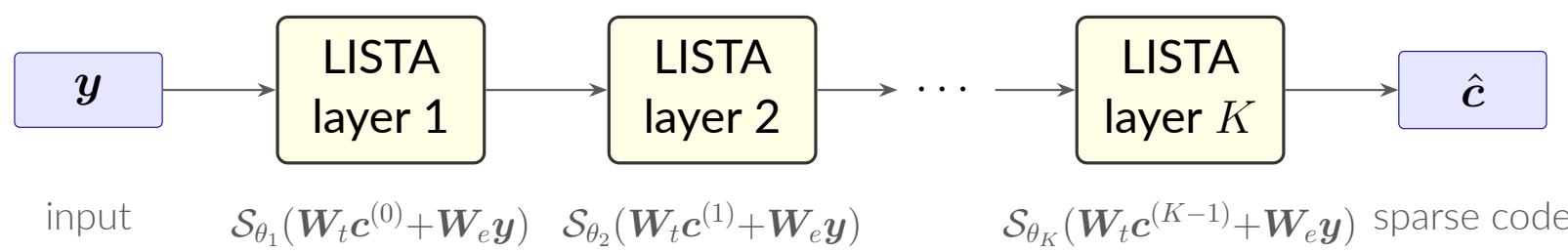


Figure 2. K -layer LISTA with tied weights $\boldsymbol{W}_c, \boldsymbol{W}_t$ and per-layer learnable thresholds θ_k . Each block is one ISTA-style update followed by soft-thresholding; the recurrence reproduces ISTA at initialisation.

Stability tricks

Naive Adam at 5×10^{-3} produces a clean descent to 0.12 NMSE, then blows up to $\sim 10^3$. Diagnosing, $\rho(\boldsymbol{W}_t) > 1$ along some eigendirection causes the recurrence to diverge. The fix combines (i) lowering $\text{lr} \rightarrow 10^{-3}$, (ii) gradient clip-norm 1.0, and (iii) restoring the best-validation checkpoint.

Phase Transition (Sec. 3.2)

Empirical Donoho–Tanner boundary on synthetic Gaussian CS, $N=200$, 20 trials/cell, $\text{NMSE} < 10^{-3}$.

$\delta = M/N$	0.30	0.50	0.70	0.90
OMP ρ^*	0.35	0.40	0.40	0.50
FISTA ρ^*	0.20	0.30	0.50	0.65

OMP wins at low δ because exact LS-on-support beats ℓ_1 when measurements are scarce, whereas **FISTA wins at high δ** because ℓ_1 keeps expanding the feasible region while OMP plateaus on dense supports. The crossover sits near $\delta \approx 0.65$.

Rate-Distortion (Sec. 3.3)

5-image natural-scene set, SNR=30 dB, 16×16 blocks. Mean PSNR $\pm$ std across scenes.

δ	0.20	0.40	0.50	0.70
OMP	17.4 $\pm$ 4.3	20.4 $\pm$ 3.7	21.7 $\pm$ 3.2	24.9 $\pm$ 3.0
FISTA	19.4$\pm$4.5	23.6$\pm$4.1	25.1$\pm$3.9	27.6$\pm$3.5
ADMM	17.6 $\pm$ 4.8	23.5 $\pm$ 4.4	25.0 $\pm$ 4.0	27.5 $\pm$ 3.7

Verdict. FISTA and ADMM agree within 0.5 dB above $\delta=0.2$ (same Lasso objective); both beat OMP by 2–3 dB. Wide std reflects scene difficulty: moon is easy (> 30 dB), coins is hard ($\sim$ 20 dB).

LISTA vs. Classical at Matched Compute

Setup. $K=10$ tied-weight LISTA trained on 5000 synthetic $(\boldsymbol{y}, \boldsymbol{c})$ pairs with $N=200$, $M=80$, $S=10$, and SNR=30 dB.

Solver	Iters/Layers	Val NMSE
ISTA	10	0.467
FISTA	10	0.337
LISTA	10	0.050
FISTA (matched)	22	$\approx$ 0.05
FISTA (conv.)	500	0.005

2.2 $\times$ **inference speedup** at matched recovery; LISTA still trails the converged ℓ_1 minimum by 10 $\times$, but amortises that gap into a 10 $\times$ smaller cost at deployment.

Joint vs. Sequential: Per-Scene PSNR Gain (dB)

8 $\times$ horizontal illumination gradient, SNR=25 dB, 16×16 center patch per scene. $\Delta = \text{PSNR}_{\text{joint}} - \text{PSNR}_{\text{seq}}$.



Figure 3. Per-scene PSNR gain over the sequential baseline. Textured wins +1.1 to +4.7 dB on cameraman, astronaut; **moon collapses by $\sim$ 8 dB** – a real, mechanism-explainable failure mode.

Why Moon Fails & Limitations

Moon’s centre patch is mostly black sky after 8 $\times$ gradient multiplication, so $\boldsymbol{B}^\top \boldsymbol{B}$ becomes rank-deficient and per-column gains are non-identifiable; the alternation then drives $\boldsymbol{g}$ to a noise-fitting solution. The sequential pipeline avoids this by *never* estimating $\boldsymbol{g}$ – its column-mean divisor is essentially a copy of the gradient. **When the scene cannot identify $\boldsymbol{g}$, refusing to estimate it is a virtue.** **Deployment rule:** gate the joint solution on $\kappa(\boldsymbol{B}^\top \boldsymbol{B}) < \tau$ and fall back to sequential otherwise.

Other limitations. The 1D illumination model suits horizontal gradients (2D vignettes need a richer prior); synthetic-trained LISTA needs distribution-wide noise training; coded-aperture validation and outer-loop unfolding are future work.

References